

CSS142P - MODELLING AND SIMULATION THEORY
FINAL PROJECT PRESENTATION



Priority-Tier Queueing in an LLM API Inference Service

Presented by: Group 6 | AM1

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The Problem and Decision Question

The Problem:

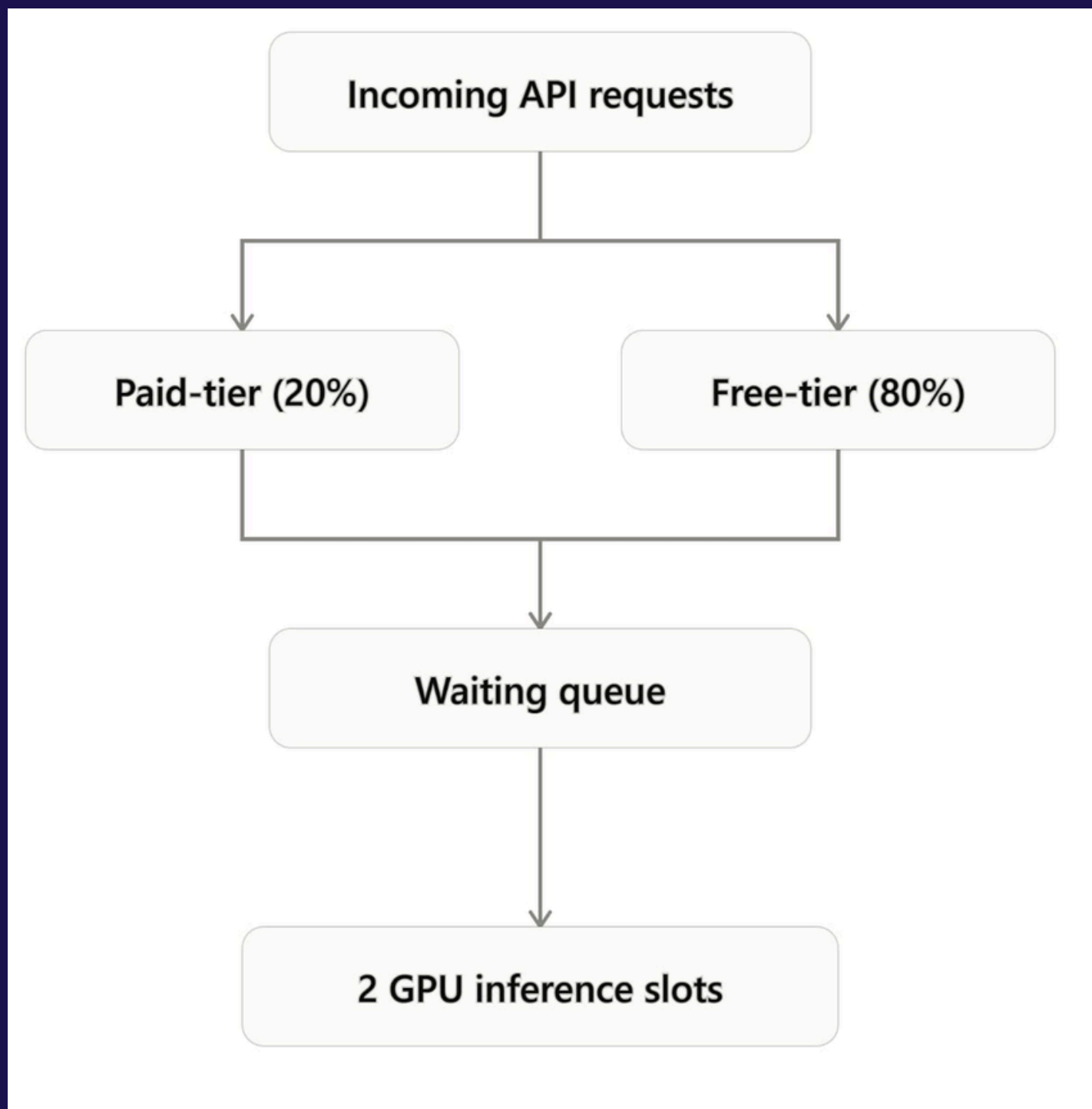
Current 2-GPU inference system uses First-Come, First-Served (FCFS) routing. During peak hours (8 req/min), Free-tier traffic bottlenecks the system, risking our 100-second Service Level Agreement (SLA) for premium Paid-tier users.

The Goal:

Evaluate a software solution (Priority Queue) versus a hardware solution (Adding a 3rd GPU) to eliminate SLA breaches.



CONCEPTUAL MODEL



- Probabilistic split: 20% Paid, 80% Free.
- Shared waiting queue.
- 2 concurrent GPU inference slots.

Key Assumptions

Arrivals:

Poisson process (3 req/min off-peak, 8 req/min peak).

Peak Windows:

9 AM–12 PM and 6 PM–9 PM (Diurnal traffic).

Service Time:

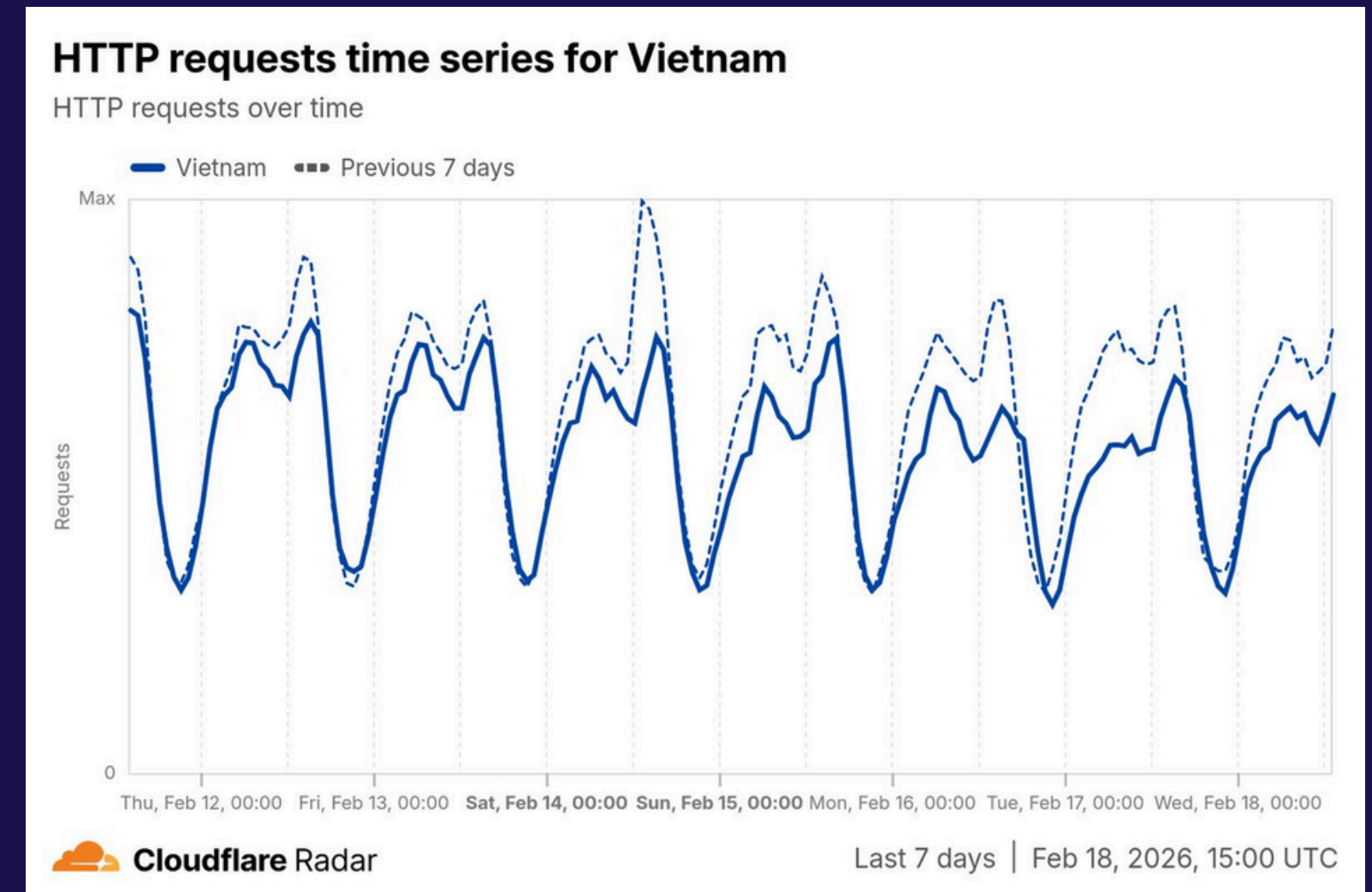
11 seconds mean (Exponential).

Customer Base:

20% Paid / 80% Free (Based on API tier data).

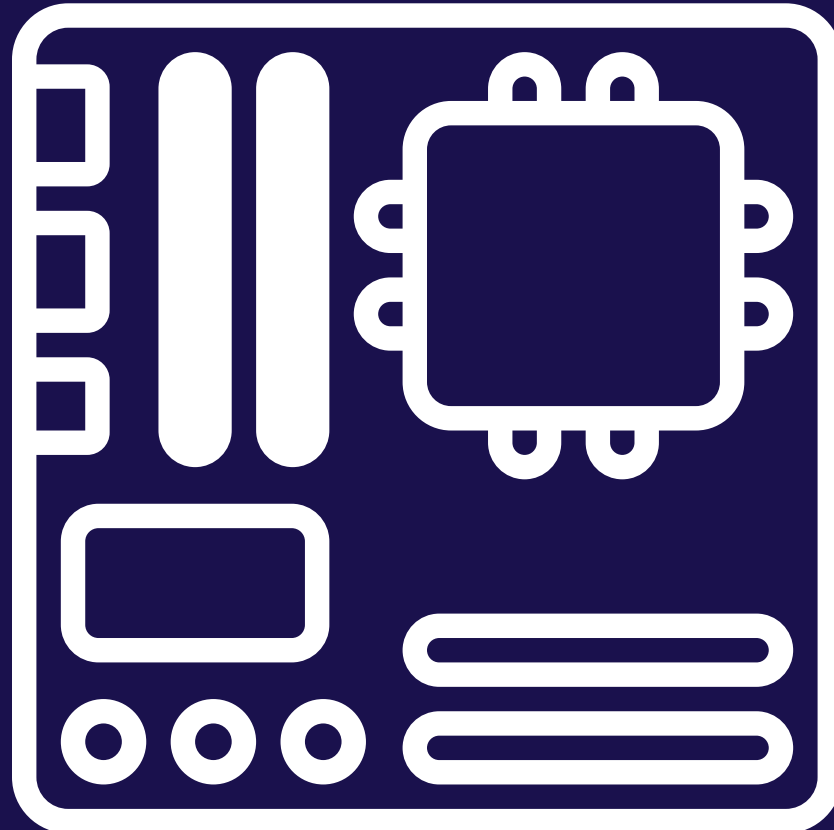
Failure Threshold:

>100 seconds queue wait (SLA Breach).

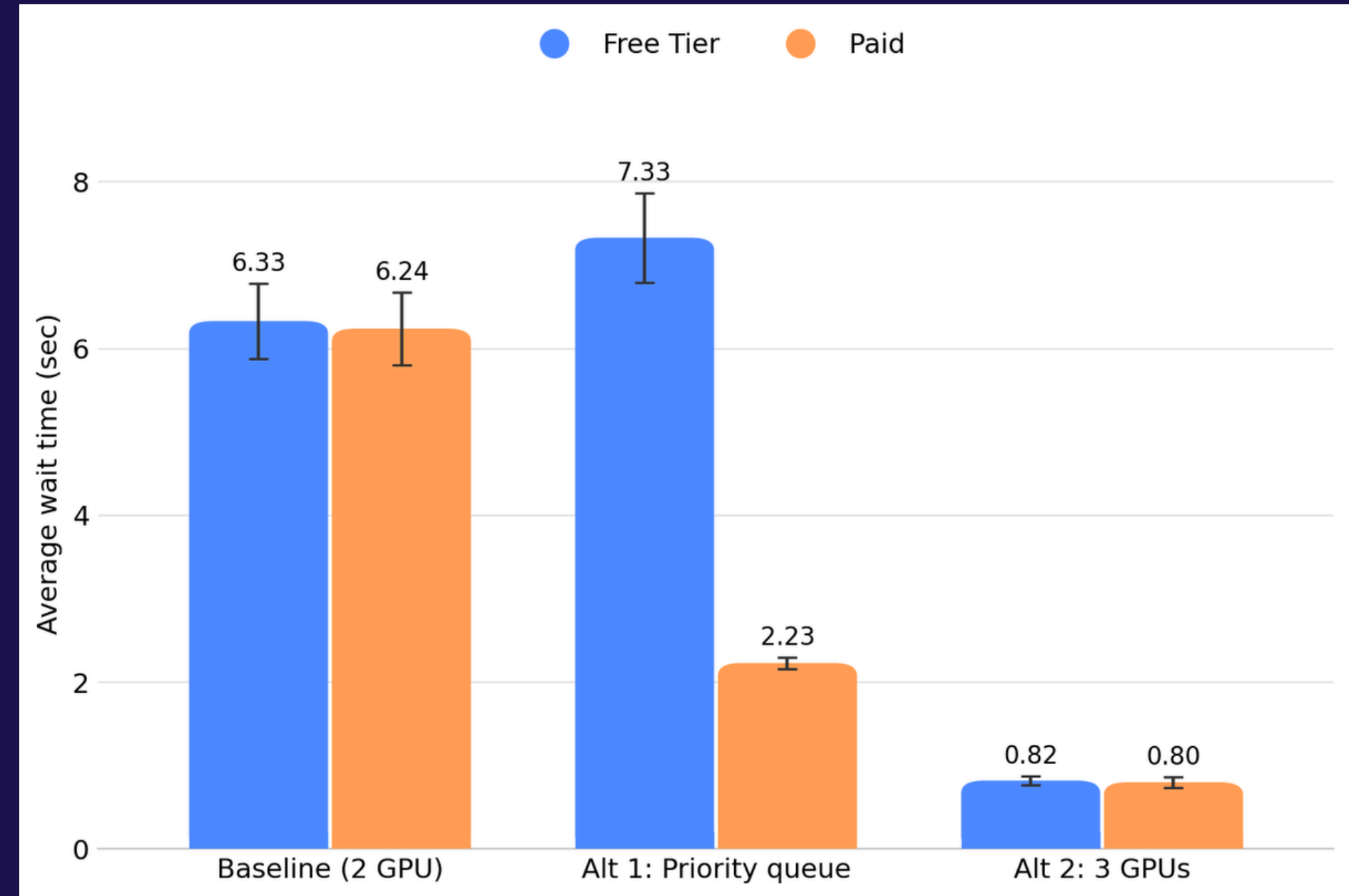


Experimental Scenarios

SCENARIO 1	SCENARIO 2	SCENARIO 3
<u>BASELINE</u> 2 GPUs FCFS <i>(Current State)</i>	<u>ALTERNATIVE 1</u> 2 GPUs Priority Queue <i>(Software Fix - Paid users skip the line)</i>	<u>ALTERNATIVE 2</u> 3 GPUs FCFS <i>(Hardware Fix - Increased server capacity)</i>



Simulation Results (Normal Traffic)



- **Baseline (2 GPUs):** Paid tier wait time is 6.24s with a 0.13% SLA breach rate.
- **Alternative 1 (Priority Queue):** Paid tier wait time drops significantly to 2.23s, effectively eliminating SLA breaches (0.00%). Free tier sees a minimal increase to 7.33s.
- **Alternative 2 (3 GPUs):** Eliminates capacity constraints entirely, dropping wait times for both tiers to under 1 second.

Sensitivity Analysis (+20% Stress Test)

Baseline Failure:

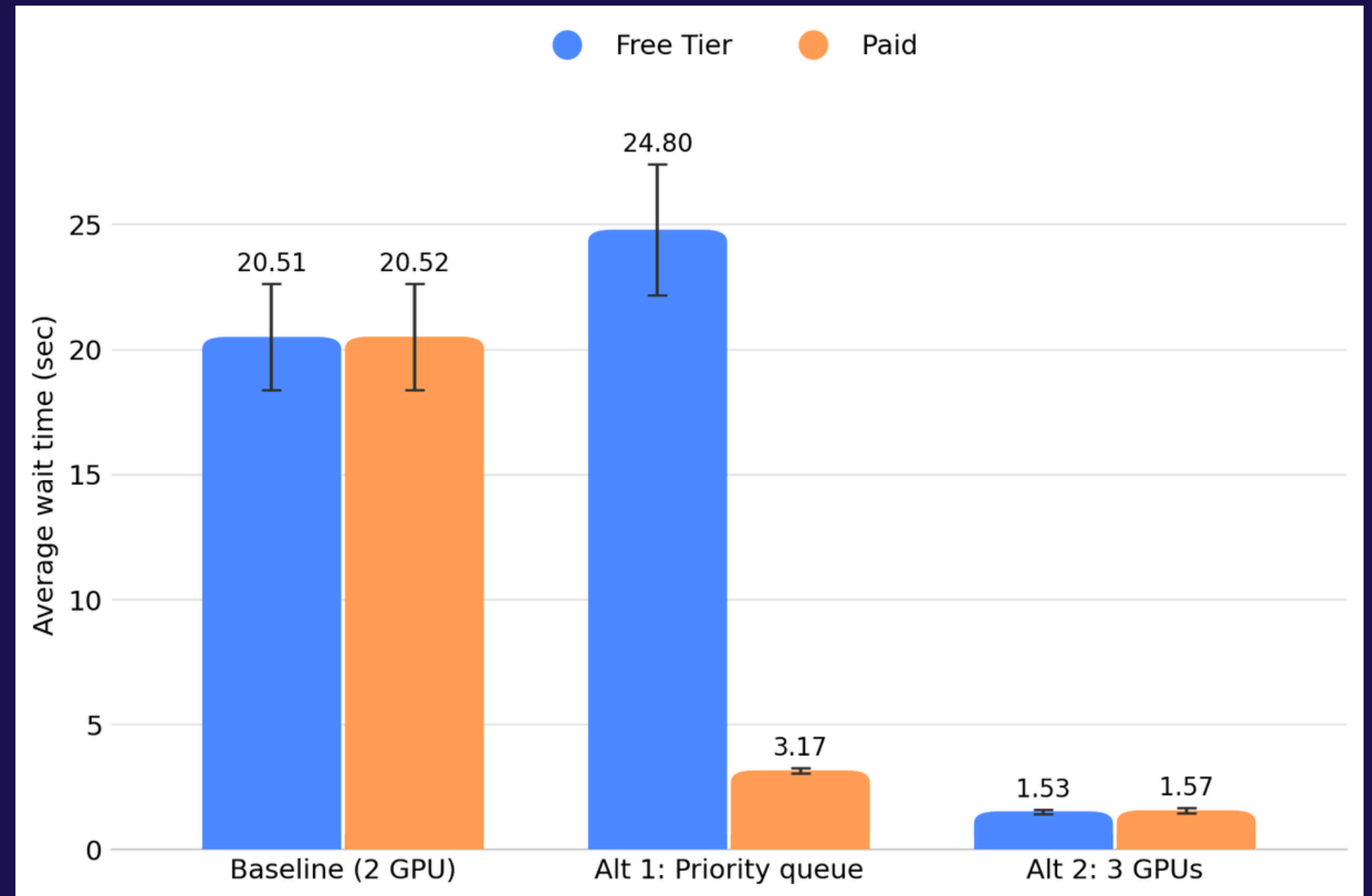
A modest 20% traffic surge causes the Baseline to crash, tripling the Paid-tier wait time to 20.52s.

Alternative 1 Resilience:

Absorbs the extra load easily; Paid-tier wait time remains safe at 3.17s.

Alternative 2 Resilience:

Barely affected by the surge, maintaining a 1.57s wait time.



FINAL RECOMMENDATION

Implement
Alternative 1 (Priority Queue).

It successfully eliminates SLA breaches and survives demand spikes without the capital expenditure required to purchase a 3rd physical GPU.

SIMULATION LIMITATIONS

- Our simulation assumes every single AI request takes an average of 11 seconds to process, but in the real world, the actual time varies heavily depending on how long or complex the user's prompt is.
- We also assumed that every user will wait patiently in line no matter what. In reality, if a customer is forced to wait past the 100-second limit, they would likely just cancel their request and leave, which would naturally change how the queue behaves.
- Finally, the model imagines a perfect world where the technology never breaks. We did not factor in random, real-world tech issues like hardware crashes, slot outages, or unexpected server downtime.